



From *in silico* to *in spectro* kinetics of respiratory complex I[☆]

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ABSTRACT

An enzyme's activity is the consequence of its structure. The stochastic approach we developed to study the functioning of the respiratory complexes is based upon their 3D structure and their physical and chemical properties. Consequently it should predict their kinetic properties. In this paper we compare the predictions of our stochastic model derived for the complex I with a number of experiments performed with a large range of complex I substrates and products. A good fit was found between the experiments and the prediction of our stochastic approach. We show that, due to the spatial separation of the two half redox reactions (NADH/NAD and Q/QH₂), the kinetics cannot necessarily obey a simple mechanism (ordered or ping-pong for instance). A plateau in the kinetics is observed at high substrates concentrations, well evidenced in the double reciprocal plots, which is explained by the limiting rate of quinone reduction as compared with the oxidation of NADH at the other end of complex I. Moreover, we show that the set of the seven redox reactions in between the two half redox reactions (NADH/NAD and Q/QH₂) acts as an electron buffer. An inhibition of complex I activity by quinone is observed at high concentration of this molecule, which cannot be explained by a simple stochastic model based on the known structure. We hypothesize that the distance between the catalytic site close to N2 (iron/sulfur redox center that transfers electrons to quinone) and the membrane forces the quinone/quinol to take several positions in between these sites. We represent these possible positions by an extra site necessarily occupied by the quinone/quinol molecules on their way to the redox site. With this hypothesis, we are able to fit the kinetic experiments over a large range of substrates and products concentrations. The slow rate constants derived for the transition between the two sites could be an indication of a conformational change of the enzyme during the quinone/quinol movement. This article is part of a Special Issue entitled: 17th European Bioenergetics Conference (EBEC 2012).

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1. Introduction

The NADH-ubiquinone oxidoreductase (complex I, EC: 1.6.5.3) catalyzes the following reaction:



H⁺_{in} depicts the protons inside the matrix, while H⁺_{out} is the protons released in the intermembrane space. The energy gained in this reaction is converted in an outward translocation of protons, with an H⁺/2e[−] stoichiometry of 4 [1–4] or perhaps 3 [5]. The kinetic mechanism of this reaction is controversial. Fato et al. (1996) [6] proposed a ping-pong mechanism in the case of mitochondria isolated from bovine heart with the oxidation of NADH preceding the reduction of ubiquinone. Nakashima et al. [7] used CoQ₁ as the electron

acceptor to analyze the activity of complex I purified from bovine heart. They proposed an ordered sequential mechanism with CoQ₁ binding as the first step and CoQ₁H₂ releasing as the last step. Hano et al. [8] also assumed that the kinetics of complex I obeys an ordered sequential mechanism when they used decylubiquinone (DQ) as the electron acceptor.

A strong inhibition, that could impede the spectrophotometric assays, is observed by the product decylubiquinol (DQH₂) particularly in the beef heart enzyme [9]. More unexpected is an inhibition by the substrate Q at high concentrations (around 100–150 μM) (Nakashima 2002b) [10]. To explain these results a second binding site for quinone was proposed by these authors and this hypothesis encountered some confirmation with the measurement of the electron spin relaxation profiles by T. Ohnishi and her collaborators who proposed the existence of two distinct semiquinone species, namely a fast relaxing SQN_f and a slow relaxing SQN_s [11,12] separated by 12 Å.

However a second quinone binding site was not evidenced by the recent crystallographic structures [13–15]. Nevertheless, these structures show the long electron-path between the site of oxidation of NADH towards the site of reduction of Q inside the hydrophilic arm

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of complex I (see Fig. 1 and [13–19]). The position of the redox clusters delineates a clear continuous path involving at least 9 redox reactions in between the site of NADH oxidation and the site of Q reduction [13]. Thus, the situation in complex I is rather different from a classical enzymatic mechanism in which the sites of the two coupled reactions are close together or even overlap. For this reason a classical enzymatic mechanism should not be expected for complex I.

In addition, two salient features of the Q binding/reaction site are evidenced by the crystallographic structures and most likely influence the kinetic mechanism. The first one is the structure of the quinone binding site [14,18,20]. It was suggested earlier that the quinone binding pocket is a broad cavity formed by the PSST (subunit hosting center N2) and the 49-kDa subunit [21,22]. It is also the binding site of inhibitors which have been grouped in 2 or 3 classes [23–25] that bind to different but partially overlapping sites within a single large quinone binding pocket as proposed by Okun and coworkers [26]. “In addition, the mutagenesis studies also indicate the approximate location of the binding sites of different complex I inhibitor classes within the large quinone and inhibitor binding cavity at the interface of the 49-kDa and the PSST subunit” [18]. In the case of complex I from the yeast *Yarrowia lipolytica*, mutational studies demonstrate that “most residues that are critical for complex I activity form a path reaching from the N-terminal three stranded β -sheet into the cavity towards iron–sulfur cluster N2” [18]. To sum up, the quinone binding/reaction site close to the N2 iron–sulfur cluster is broad with overlapping binding (sub)sites.

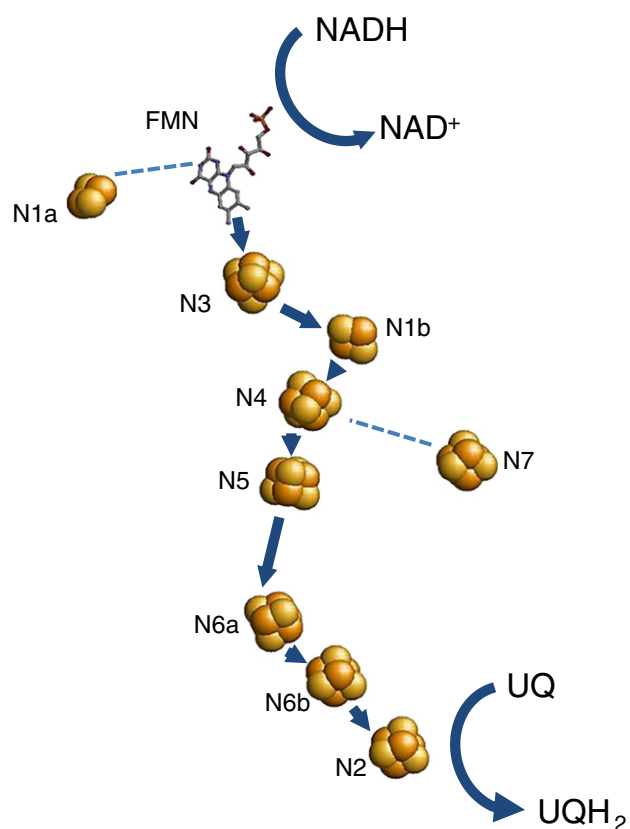


Fig. 1. Spatial location of the reactions leading from the oxidation of NADH to the reduction of ubiquinone in the hydrophilic arm of complex I (hydrophilic domain of respiratory complex I from *Thermus thermophilus* (PDB ID: 2FUG)). The full arrows indicate the main electron paths. The dotted lines describe electron paths taken to a lesser extent. All reactions are reversible so that electrons flitter back and forth between the reaction centers. The net reactions illustrated by the arrows are the result of all individual forward and backward reactions.

The second salient feature concerns the localisation of this broad quinone binding site. It is a matter of controversy. “Brandt and co-workers contend that the whole domain is ‘lifted away’ from the membrane, placing cluster 4Fe[PS]/N2 35–60 Å away from the membrane interface, and requiring quinone to move substantially out of the membrane for catalysis.... In contrast, Sazanov contend that the hydrophilic domain is positioned with 4Fe[PS]/N2 close to the membrane so that the quinone headgroup can approach 4Fe[PS]/N2, but remain in the membrane interface” [16]. More recently the group of Sazanov described a quinone site situated about 25 Å away from the membrane [27]. Thus, there could (must) exist an access path to the quinone site which is more or less long depending on the localisation of the site with regard to the membrane. Both properties have to be incorporated in any kinetic model.

There is another important question concerning complex I functioning which is not *a priori* taken into account in this study: the nature of the molecular mechanisms by which complex I couples the redox reactions, localized in the hydrophilic arm, to the vectorial proton pumping situated in the membranous arm. It most likely involves long range conformational changes in the membrane domain driven by the quinone reduction in the peripheral domain to couple the redox reactions to the proton translocation (see the discussions in [18] and [27]). Vinogradov [28] evidenced the existence of an active and an inactive conformation of the enzyme and characterized the transition occurring between them in the presence or absence of the electrochemical gradient $\Delta\mu\text{H}^+$. Thus, despite the fact that we do not explicitly take into account a conformational change, such a process should have an influence on the kinetic mechanism that becomes apparent in our model under the form of low rate constants as well as under the form of an unfitting thermodynamical balance.

We have recently developed a stochastic model [29], based on Gillespie algorithm [30] of the complex I electrons transport arising from NADH oxidation and ending with ubiquinone reduction. Our simulations showed that the two NADH electrons most of the time successively follow the same “linear” route defined by NADH⁺ site–FMN–N3–N1b–N4–N5–N6a–N6b–N2–UQ site with very little storage of the “second” electron on N1a contrary to what had been proposed in [13]. Rather, N1a acts as an acceptor or donor of electron for flavine-semiquinone, minimizing its lifetime and thus ROS production. The interest of our stochastic model is that it takes into account the structure *i.e.* the distances between the redox centers and their mid-point potentials. Another advantage of a stochastic model is that it easily allows the reproduction *in silico* of the experimental enzymatic kinetics. Indeed it is possible to repeat simulations in the same conditions (at given substrates/products concentrations) and to take the mean of the initial velocities, which is equivalent to what occurs in a spectrophotometer cuvette which actually averages individual reactions.

In this paper, we show that the *in silico* data can fit the experimental ones at low substrates concentrations and in the absence of products. The *in silico* simulations produce something which is not an ordered mechanism but is close to a ping-pong model from which it diverges at moderate and high NADH concentrations (> 15 μM), in accordance with the complexity of the complex I reaction structure and the experimental results. However our model does not fit the experimental results when the substrate inhibition is observed with oxidized ubiquinone Q. It presumably means that what occurs in the Q reducing site is more complex than a simple binding followed by the transfer of two electrons between N2 and the quinone molecule. In order to model this behavior we took into account in our model the two main features stipulated above, *i.e.* the broad cavity that accepts the quinone molecules with the existence of a more or less long path for ubiquinone/ubiquinol to access the reaction site. This indicates a movement of the quinone/quinol which necessarily occupies several positions in between the membrane and the catalytic site. We introduced these several localizations in our model under

the form of a second intermediate site for ubiquinone. It does not mean that there actually exists such a second specific site but rather that, at high concentration, quinone/quinol can obstruct the reaction site or a path conducting to it.

Our model makes it possible to fit the experimental results; furthermore the slow passage of the quinone/quinol between the two sites may be explain by a change of conformation in complex I, necessary for proton pumping.

2. Materials, methods and models

2.1. Preparation of mitochondria

Frozen isolated beef heart mitochondria were a gift from Dr. Joel Lunardi, prepared according to [31]. The final mitochondrial pellet was homogenized at a concentration of 40 mg/mL and the mitochondrial suspension was frozen in liquid nitrogen as small beads of 50 μ L volume and stored at -80°C . One bead was rapidly thawed before each experiment.

All assays were performed in temperature-controlled single wavelength spectrophotometer.

2.2. Complex I (NADH–ubiquinone oxidoreductase) assay

The assay was performed at 37°C according to [32] by following the decrease in absorbance at 340 nm resulting from the oxidation of NADH in 1 mL of medium containing 65 mM KH_2PO_4 (pH 7.5), 2 mg BSA, 2 mM EDTA, 46 μ M antimycin A, 4.4 μ g mitochondrial protein and in control assays additionally 25 μ M rotenone. The concentrations of the substrates NADH and Q (decylubiquinone) varied, as well as the concentrations of products NAD^+ and QH_2 . The reaction was initiated by NADH addition. The extinction coefficient used for NADH concentration determination was $6.22 \text{ mM}^{-1} \text{ cm}^{-1}$ at 340 nm. The net activities have been obtained by subtracting the residual activity in the presence of rotenone from the activities without rotenone. All products have been purchased by Sigma Aldrich.

2.3. Short description of the model. In silico simulations

The stochastic model used in this study is extensively described in [29]. It is based on the calculation of the rate constants of electron tunneling, k_{et} , from which we derive the reaction probabilities. These rates of electron tunneling are calculated according to the empirical Marcus-type equation [33–36]:

$$\log k_{\text{et}} = 13 - (1.2 - 0.8\rho)(D - 3.6) - \gamma \frac{(\Delta G^0 + \lambda)^2}{\lambda} \quad (2)$$

where D the distances edge to edge between the redox centers and the binding sites of complex I structure, ΔG^0 are the standard free Gibbs energies (Table 1), ρ , the packing density, is around 0.76 in a typical protein [35]. For λ , the reorganization energy, a value of 0.7 eV seems to be an adequate generic value [35], and $\gamma = F/(4RT \ln(10)) = 4.06 \text{ eV}^{-1}$ at 37°C is derived from the classical Marcus expression [33–36].

We measured the distance D on *Thermus thermophilus* structure of complex I [13]. Because the rate constants k_{et} decreases rapidly with the distance D according to Eq. (2), only transfer reactions between redox centers less than 23 Å apart are considered in our model, i.e. 19 electron (reversible) transfer reactions. The ΔG^0 values used as well as the distance D are shown in Table 2, which is adapted from [29]. Most of these values are the values collected in [33] and [37] in which we introduce a slight “rollercoaster” shape as proposed in [38]. Only one complex I molecule is used for the simulation.

Table 1

Single-electron redox center midpoint potentials in complex I after [29].

Redox couple	E^0 (in eV) modified after [29]
NADH/ NAD^+	−0.320
FMNH ₂ /FMNH•	−0.293
FMNH•/FMN	−0.389
$\text{QH}_2/\text{SQ}^{\bullet}$	0
$\text{SQ}^{\bullet}/\text{Q}$	−0.12
N1a	−0.38
N3	−0.25
N1b	−0.20
N4	−0.30
N5	−0.20
N6a	−0.30
N6b	−0.20
N2	−0.10
N7	−0.25

The simulations were done according to Gillespie algorithm [30] as described in [29].

In order to compare the simulations and the experiments, the concentrations used in the experiments, were converted into a number of molecules (used in simulations) taking into account the volume around a single complex I molecule as calculated in [29]. The experimental data are expressed in nmol/min/mg and the simulation data are in molecule/s/cplxI. To superpose both data, we calculated a conversion coefficient which is the complex I amount (nmol) in 1 mg of mitochondrial protein. It is obtained by dividing the experimentally observed V_{max} (1920 nmol/min/mg) by the catalytic constant estimated by simulation at high substrates concentrations (292 s^{-1}). The conversion coefficient is thus 6.6, corresponding to an enzyme concentration of 0.48 nM. Various numbers of Q, QH_2 , NADH and NAD^+ molecules were used corresponding to the different concentrations in the kinetic experiments.

2.4. Fitting procedure

For the original model 1 (1 Q site) only the k_{on} and k_{off} of the binding and release of substrates and products have to be fitted. The other rate constants are imposed by the structure of the complex I and its physical chemistry.

The fitting procedure is, in part, performed manually: a set of values for the binding/release rate constants (Tables 3–4) and additionally, in the 2 Q sites model 2, a set of values for the rate constant of the transition between the Q sites (Table 5) are chosen. For each experimental curve several simulations were performed with the set of parameters values. Each curve is a function of a given substrates/products concentrations, the other parameters being fixed. Each simulation corresponds to 3 s of enzymatic reaction which is enough to reach and maintain a steady state and measure an initial velocity to be compared with the experimental results. For that purpose, we used a hyperbolic fit of the experimental points with or without Q inhibition because the substrates concentrations of the simulated and experimental curves were not always the same. The mean of the simulated rate for each condition (v_{sim}) is compared to the corresponding point on the hyperbolic fit of the experimental results in the same condition (v_{exp}). The sum of the square of differences $C = \sum (v_{\text{exp}} - v_{\text{sim}})^2$ is taken as a measure of the fit's accuracy. The parameters are refined in order to minimize C. It should be duly noted that, due to the stochastic nature of the process, C is a random variable prone to random variations for the same set of parameters. The experimental points are plotted as points together with the mean of the initial rate of stochastic simulations in the same conditions represented as continuous curves in order to discriminate them from the experimental points (Figs. 2 and 5).

3. Results

3.1. Implication of the complex I structure to the kinetic mechanism

Fig. 1 shows that the reaction catalyzed by complex I must be rather different than a simple coupling between two redox reactions. The main feature is that the two half redox reactions are widely separated (around 90 Å) by a series of redox reactions, so that the reduction of ubiquinone at one end appears physically independent from the oxidation of NADH at the other end of the enzyme. It *a priori* rules out an order in the binding of NADH and Q and of the release of NAD and QH₂ (ordered mechanism) or an order in the release of a product before the binding of the other substrate (ping pong mechanism). This does not mean that an ordered mechanism does not occasionally exist but rather that if an order exist it is just aleatory or the consequence of a special arrangement of rate constants and concentrations. In our previous stochastic simulations based on the Gillespie algorithm [29] we showed that most of the electrons take the same route defined by the successive sites and redox centers: NADH⁺ site-FMN-N3-N1b-N4-N5-N6a-N6b-N2-Q site, which involves seven intermediate reactions between the oxidation of NADH and the reduction of Q (nine reactions all together). The simulations in [29] based on the structure and properties of the hydrophilic arm of complex I concern one single molecule of complex I. The mean of several simulations for the same given concentrations of substrates and products mimics the experimental kinetics as recorded in a spectrophotometer, where a great number of complex I molecules are present. We thus addressed the following questions: 1— what kind of kinetic mechanism do we obtain in these simulations? 2— how do the simulations relate to the experimental results?

3.2. Kinetics at low substrates concentrations

The experimental initial velocities at different concentrations of NADH and Q (in the absence of the products NAD and QH₂) were plotted (as points) together with the mean of the initial rate of ten stochastic simulations in the same conditions (represented as continuous curves in order to discriminate them from the experimental points, Fig. 2). The k_{on} , k_{off} and K_d values are summarized in Table 3.

Table 2

The distances between redox centers are from the 3D structures (hydrophilic domain of respiratory complex I from *Thermus thermophilus* (PDB ID: 2FUG)). ΔG^0 are calculated from the E^0 values after [28]. k_f (forward reaction) and k_b (backward reaction) are calculated as explained in the model section. QH₂, SQ• and Q mean ubiquinol, semiquinone and ubiquinone respectively.

Reactions	Distance (in Å)	ΔG^0 (in eV)	$k_{forward}$ (in s ⁻¹)	$k_{backward}$ (in s ⁻¹)
2 e ⁻ transfer from NADH to FMN	1.0	0.042	2.21 10 ¹¹	1.06 10 ¹²
e ⁻ transfer from FMNH ₂ to N1a	11.3	0.087	7.03 10 ⁴	1.82 10 ⁶
e ⁻ transfer from FMNH• to N1a	11.3	-0.009	4.68 10 ⁵	3.34 10 ⁵
e ⁻ transfer from FMNH ₂ to N3	7.6	-0.043	1.34 10 ⁸	2.68 10 ⁷
e ⁻ transfer from FMNH• to N3	7.6	-0.139	6.39 10 ⁸	3.52 10 ⁶
e ⁻ transfer from N3 to N1b	11.0	-0.05	1.47 10 ⁶	2.26 10 ⁵
e ⁻ transfer from N3 to N4	13.8	0.05	4.98 10 ³	3.23 10 ⁴
e ⁻ transfer from N1b to N4	10.7	0.10	1.21 10 ⁵	5.10 10 ⁶
e ⁻ transfer from N1b to N5	16.8	0	220	220
e ⁻ transfer from N4 to N5	8.5	-0.10	1.02 10 ⁸	2.43 10 ⁶
e ⁻ transfer from N5 to N6a	14.0	0.10	1.35 10 ³	5.68 10 ⁴
e ⁻ transfer from N6a to N6b	9.4	-0.10	3.00 10 ⁷	7.12 10 ⁵
e ⁻ transfer from N6b to N2	10.4	-0.10	7.68 10 ⁶	1.82 10 ⁵
e ⁻ transfer from N1a to N3	19.4	-0.13	577	0.45
e ⁻ transfer from N4 to N7	20.4	-0.05	4.00	0.62
e ⁻ transfer from N3 to N5	21.1	-0.05	1.54	0.24
e ⁻ transfer from N5 to N6b	22.3	0	0.12	0.12
e ⁻ transfer from N2 to Q	12	0.02	1.04 10 ⁵	2.21 10 ⁵
e ⁻ transfer from N2 to SQ•	12	-0.10	8.67 10 ⁵	2.06 10 ⁴

Table 3

Kinetic constants: rate constants of substrates/products binding and release on complex I determined by trial and error fitting of experimental kinetics with the 1Q stochastic model (only one Q site).

	k_{ON} [M ⁻¹ s ⁻¹]	k_{OFF} [s ⁻¹]	K_d [M]	E [V]
NADH	29,000,000	285	9.8 10 ⁻⁶	-0.308
NAD	2,200,000	1500	682 10 ⁻⁶	-0.195
Q	15,000,000	580	38.7 10 ⁻⁶	-0.272
QH ₂	37,500,000	320	8.5 10 ⁻⁶	-0.312

Table 4

Binding constants on Q-site 1 determined by trial and error fitting of the experimental points of the experimental kinetics represented in Fig. 5 with the stochastic 2Q sites model.

	k_{ON} [M ⁻¹ s ⁻¹]	k_{OFF} [s ⁻¹]	K_d [M]	E [V]
NADH	16,500,000	400	24 10 ⁻⁶	-0.285
NAD	2,800,000	1800	643 10 ⁻⁶	-0.198
Q	15,000,000	2700	180 10 ⁻⁶	-0.107
QH ₂	20,000,000	1200	60.10 ⁻⁶	-0.137

The simulations are in accordance with the experimental results and give kinetics which cannot be recognized as following an ordered or ping pong mechanism. It is particularly clear on the reciprocal plots (Fig. 3) and particularly at high concentrations of NADH where no linear rate dependency on 1/NADH are observed.

3.3. Kinetics at saturating NADH concentrations

At saturating NADH concentrations (Fig. 2(a) to (g)) a plateau is evidenced both with the experimental points and with the simulated curves, so that the results cannot be properly fitted by a simple hyperbola. Indeed, either the hyperbola fits the first points at low substrate concentration with a too high V_{max} or the hyperbola fits the points at high concentration of NADH but with a too low K_m . This plateau is presumably caused by a buffer effect brought in by the seven intermediate reactions in between the site of NADH oxidation and the site of Q reduction. The adequacy between the simulated and the experimental kinetics evidencing a clear plateau should be noted. It reinforces the idea that complex I kinetics does not follow a traditional kinetic mechanism as a consequence of the facts that the two half redox reactions are not only chemically different but also physically separated.

3.4. Kinetics at high concentrations of decylubiquinone

At high concentration of DQ (decylubiquinone) an unexpected inhibition of complex I activity is observed between 50 and 150 μM (see Fig. 5(q) and (r)). Although a product competitive inhibition (by NAD and DQH₂) is expected and observed (Fig. 5(s) to (x)), an inhibition by a substrate excess (DQ) is not predicted by our stochastic model based on the sites and redox centers evidenced on the crystal structure. As

Table 5

Rate constants of transfers between sites A and B in the 2Q sites model determined by trial and error fitting of the experimental kinetics represented in Fig. 5.

	$k_{A \rightarrow B}$ [s ⁻¹]	$k_{B \rightarrow A}$ [s ⁻¹]	K_{eq}
Q	500	2550	5.1
QH ₂	800	1500	1.9

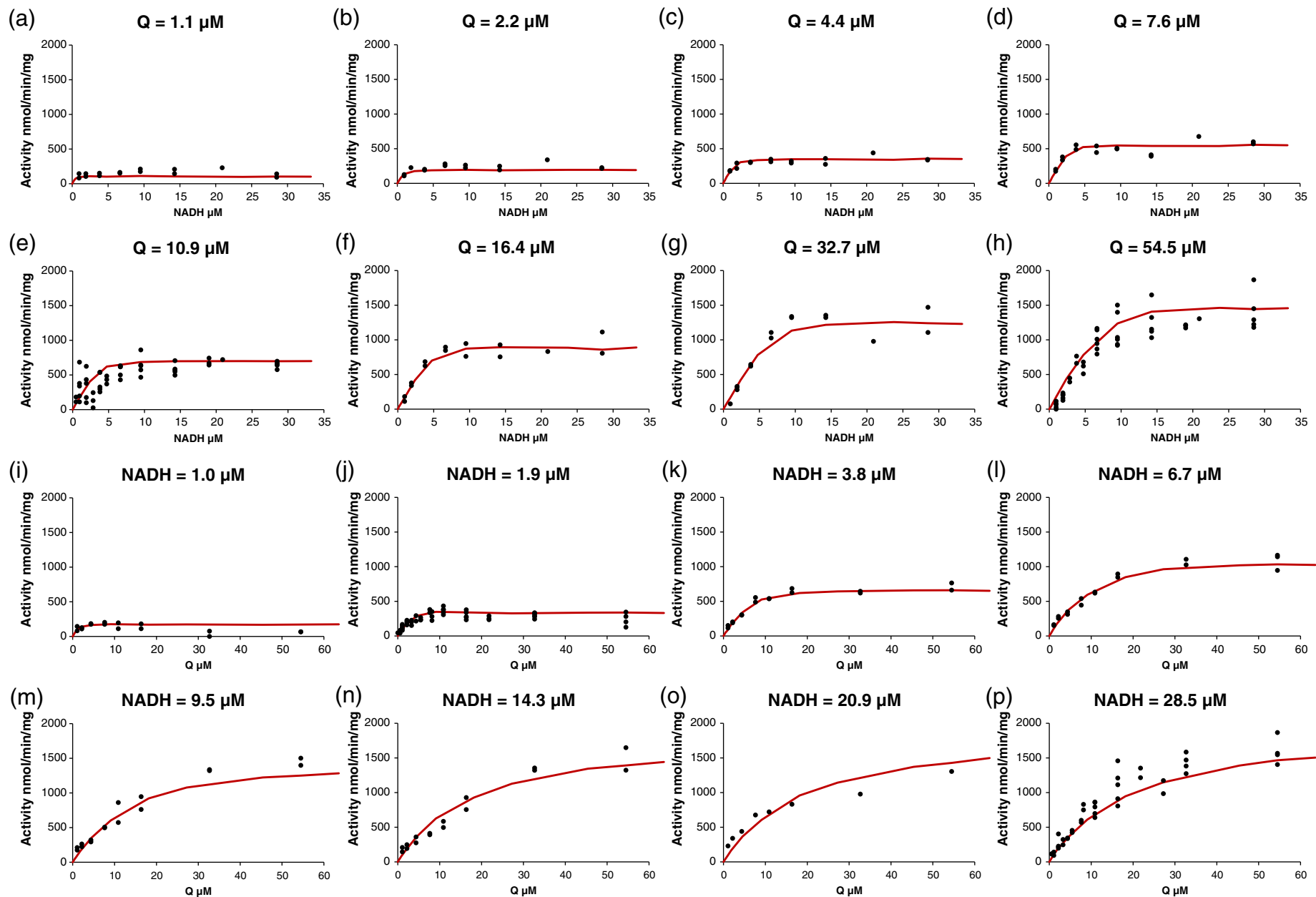


Fig. 2. Complex I kinetics at low substrates concentrations. The points are the experimental points. The curves are links between means of simulated kinetics according to the model with one Q site using parameters of Table 2 (model 1).

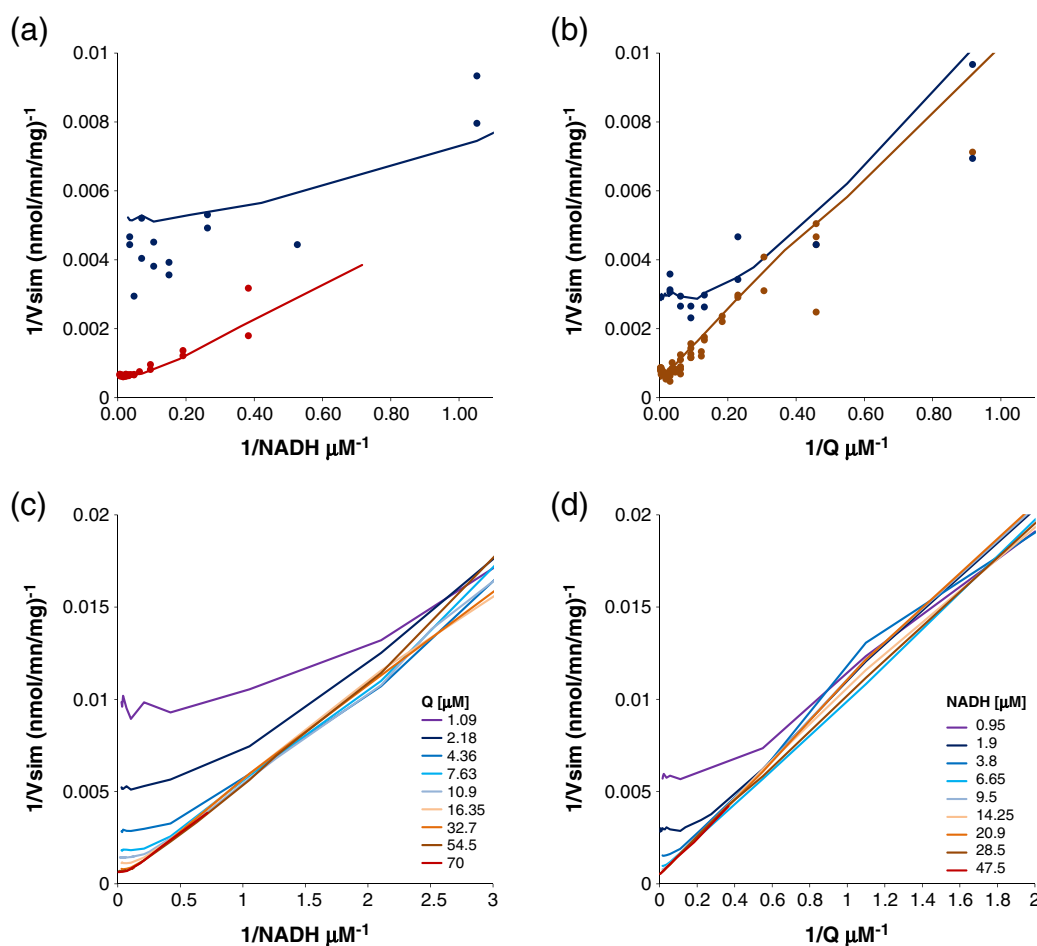


Fig. 3. Double reciprocal plots evidencing a plateau in some conditions. The points represent the experimental data and the curves are the simulated ones. (a): Double reciprocal plots of experimental kinetics as a function of $1/NADH$ at $Q = 2.18 \mu M$ and $Q = 70 \mu M$. (b): Double reciprocal plots of experimental kinetics as a function of $1/Q$ at $NADH = 1.9 \mu M$ and $NADH = 28.5 \mu M$. (c), (d): Double reciprocal plots of the simulated points.

discussed in the introduction, it led us to hypothesize a more complex mechanism at least around the quinone binding site. In accordance with the structural data that place cluster N2 more or less away from the lipid bilayer [14,17,18,27], suggesting that the hydrophobic quinone must spring out of the membrane to get its electrons from N2, we introduced in our model a second (inhibitory) quinone site, site B in addition to the quinone redox site A close to N2. This new site represents in a single place the different possible locations of quinone species in the large quinone site around N2 (site A) or in the membrane as well as the path taken by the quinone species from (to) the membrane to (from) the quinone redox site (site A close to N2) (Fig. 4A).

In this model, the new site B has to be occupied by Q on its way towards N2 or by QH_2 on its way back. At high Q concentrations this site B will be occupied most of the time and will thus inhibit QH_2 release from N2. More generally, both sites can be occupied by Q, SQ and QH_2 which leads to the mechanism represented in Fig. 4B.

We derived a stochastic model from the mechanism depicted in Fig. 4 and fitted it to a series of experimental curve representative of various conditions of substrates and products concentrations, including the experiments of Fig. 2 at low substrates concentrations. The fit was performed by adjusting the binding and release rate constants of the different sites (NAD/NADH site and site 1 for Q/ QH_2) and the transition rates of the quinone species (Q/S/ QH_2) from one site to the other ($k_{A \rightarrow B}$ and $k_{B \rightarrow A}$). As stressed in the Materials, methods and models section, the other rate constants were imposed by the structure of complex I and the midpoint potentials of the redox centers as explained in [29] (summarized in Tables 1 and 2).

A rather good fit is obtained with the values listed in Tables 4 and 5 and the corresponding curves were drawn in Fig. 5 (experimental data as points and simulations as continuous red curves).

4. Discussion

4.1. Structure and kinetics of complex I

Considering the organization of the reaction centers in the hydrophilic arm of complex I depicted in Fig. 1 it is not surprising that a classical enzymatic mechanism such as an ordered or a ping pong mechanism which are based on adjacent or even overlapping reaction sites is not obtained. The distance (around 90 Å) and the seven reactions in between the oxidation of NADH and the reduction of ubiquinone made the two sites as if they were independent from each other. It rules out any *a priori* compulsory order in the binding of substrates or in the release of products preventing the kinetics of complex I to obey an ordered or a ping-pong mechanism as reported before [6–8,10]. This is in accordance with our kinetic experiments which do not evidence a clear ordered or ping-pong mechanism (Figs. 2 and 3) particularly when studying kinetics on a large range of substrates/products concentrations.

4.2. Kinetics in silico vs kinetics in spectro

Because we have developed a stochastic model for the reactions occurring inside a complex I molecule it was tempting to use this

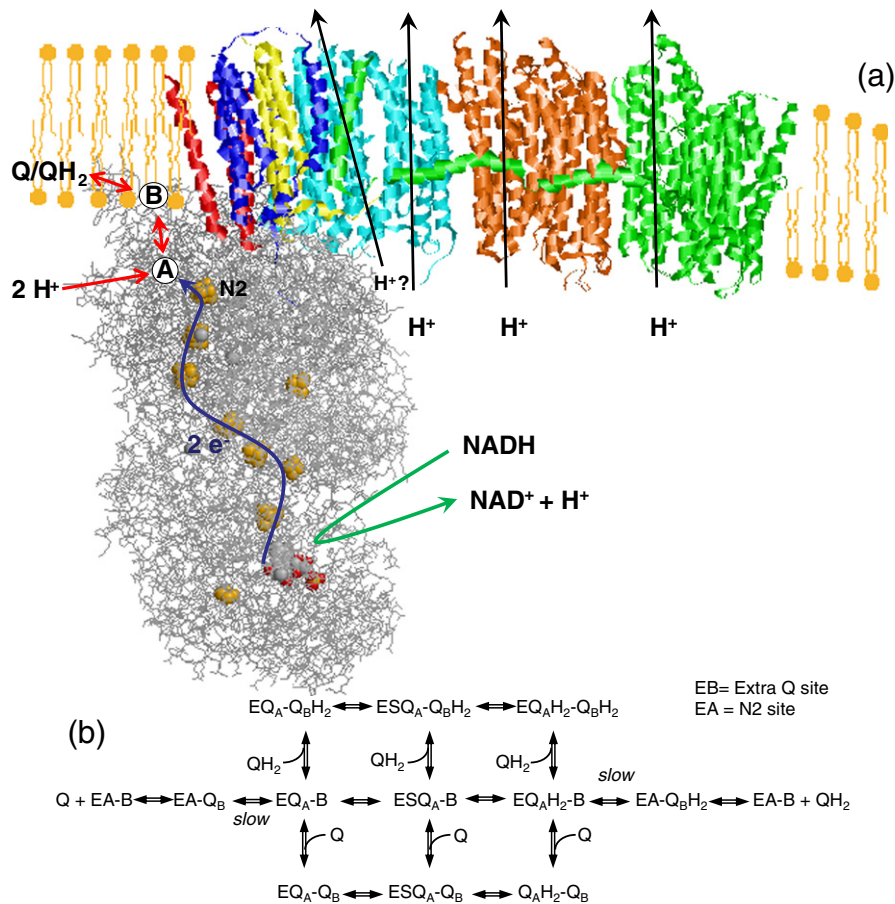


Fig. 4. Model 2Q sites. (a): Model taking into account the quinone path between the membrane and possible subsites in the large quinone site around N2 center (site A), symbolized by the supplementary site B. (b): EA-B represents the complex I with free sites A and B. The site A is the oxido-reduction site close to N2. Both sites A and B can bind the three molecules Q, SQ and QH₂ leading to species EA-Q_B, EQ_A-Q_B etc. The pathway in the middle of the figure is the “ideal” path for Q reduction, which occurs from the left to the right and back without inhibition: a Q molecule binds to site B, pass to site A, is reduced in SQ_A then in Q_AH₂, comes back to site B and is then released from complex I molecule.

model to simulate enzymatic kinetics as they are performed in a spectrophotometer. As stated above the simulation concerns only one complex I molecule. It is thus necessary to average several simulations (corresponding to the numerous complex I in a spectrophotometric cuvette averaged by the photomultiplier) to simulate the signals recorded experimentally. By doing so, it is easy to go from the molecular level to a more integrated level of enzymatic kinetics. It should be emphasized here that the simulations are partly determined by the structure (distances) and the midpoint potentials of the reaction centers. Only the binding and release rates of the substrates and products can be adjusted to the data, which corresponds to the usual way of adjusting a kinetic model to kinetic experiments. On the other hand, it must be stressed that the internal structure and the set of redox reactions inside an isolated complex I molecule impose a behavior which is independent of the adjustable binding and release rate constants of substrates and products.

Fig. 2 and Table 3 show that it is possible to find a set of binding and release rate constants that fit all the kinetic experiments at low concentrations of substrates in the absence of products. The resulting K_d for NADH (9.8 μM) and for NAD (682 μM) is in the range of the literature values. Vinogradov reported similar values for beef heart and *Paracoccus denitrificans* (2, 5 and 7 μM for NADH depending on the pH and 1000–1600 μM for NAD in the case of heart submitochondrial particles [28,39]. Nakashima et al. [7] found a K_m around 2 μM for NADH depending of Q1 and NAD concentrations. Their K_m for Q1 is around 10–18 μM at low Q1 concentration although their value is

around 500 μM at higher concentrations. In our simulations the K_d for decylubiquinone is slightly higher (38.7 μM at low decylubiquinone concentration, Table 3). This difference can be explained by the fact that the two quinone species are different. Indeed, Fato et al. [6] found similar values for NADH ($K_m = 8.7 \mu\text{M}$ and 9.2 μM) but lower values for decylubiquinone ($K_m = 1.8 \mu\text{M}$ and 2.1 μM). They show that the K_m value depends upon the type of quinone used (K_m varies from 0.8 μM to 21 μM). Furthermore the comparison of apparent K_m (depending on NADH and Q concentrations) with K_d (simple ratio of release/binding constants) is not completely allowable, even if both parameters are usually of the same order. It should be noticed that, in our model, the fitted K_d of QH₂ is lower than the K_d of Q which corresponds to the strong QH₂ inhibition observed experimentally (see Fig. 5(s) and (t)).

Because the quinone is mainly stored in the membrane, its membrane concentration is much higher than the concentration calculated using the full volume of the assay, as is done for NADH and NAD. The K_d calculated in these conditions is thus higher. This point is particularly well studied by Fato et al. in [6] who give a higher value of 138 mM.

4.3. Plateau at high concentration of NADH

Fig. 2(a) to (h) evidence a plateau for the saturating NADH concentrations used in these experiments. This plateau is well illustrated in the double reciprocal plot representation of Fig. 3A. Such a plateau

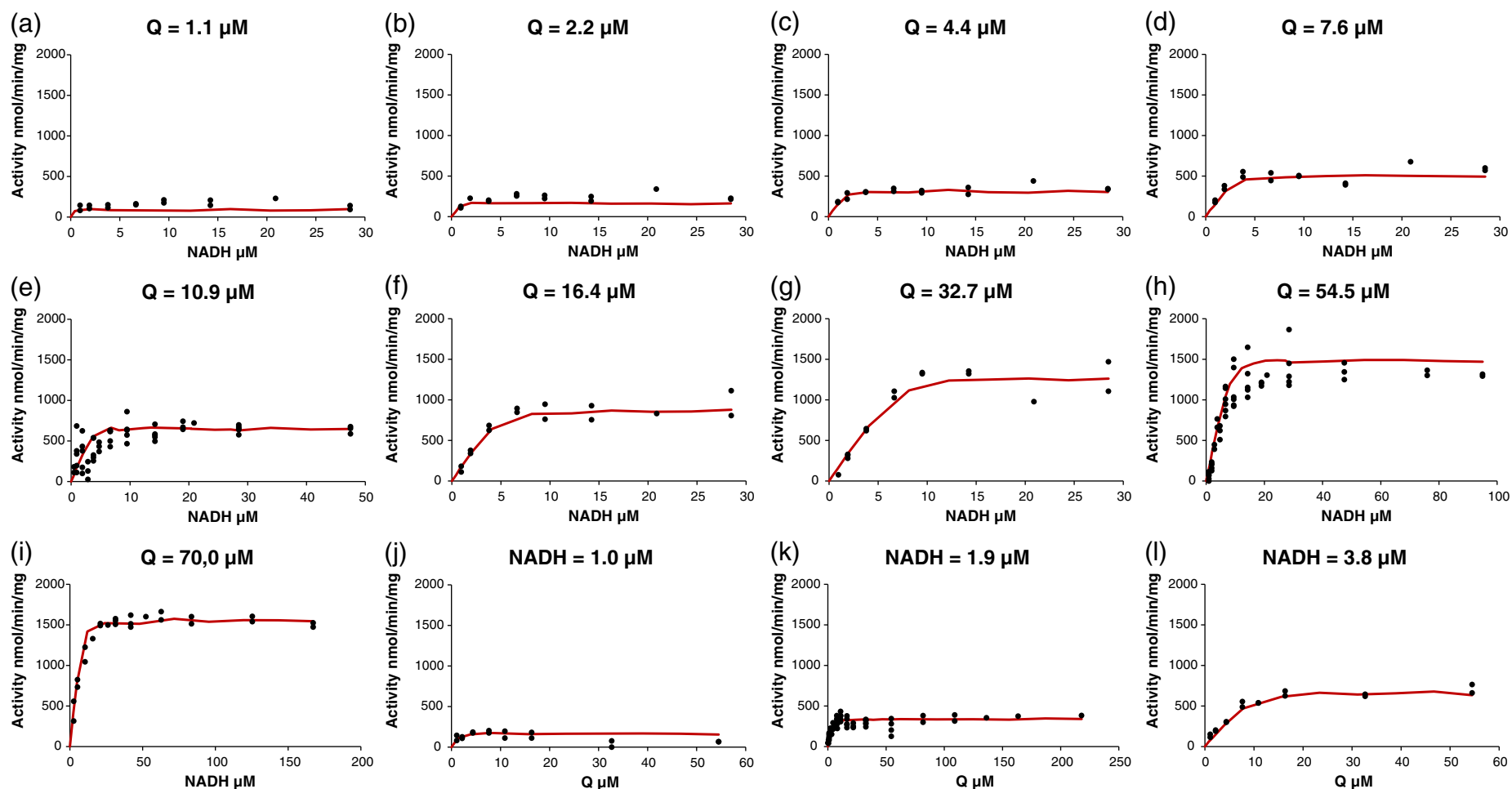


Fig. 5. Simulations of initial rates at low and high substrate and product concentrations with the 2 quinone sites model 2. Points represent experimental data. Curves represent the simulations performed with the two quinones stochastic model 2 represented on Fig. 4 according to the values in Tables 4 and 5. (x): rate as a function of QH_2 with $Q + QH_2 = 56.8 \mu M$ ($Q + QH_2$ constant).

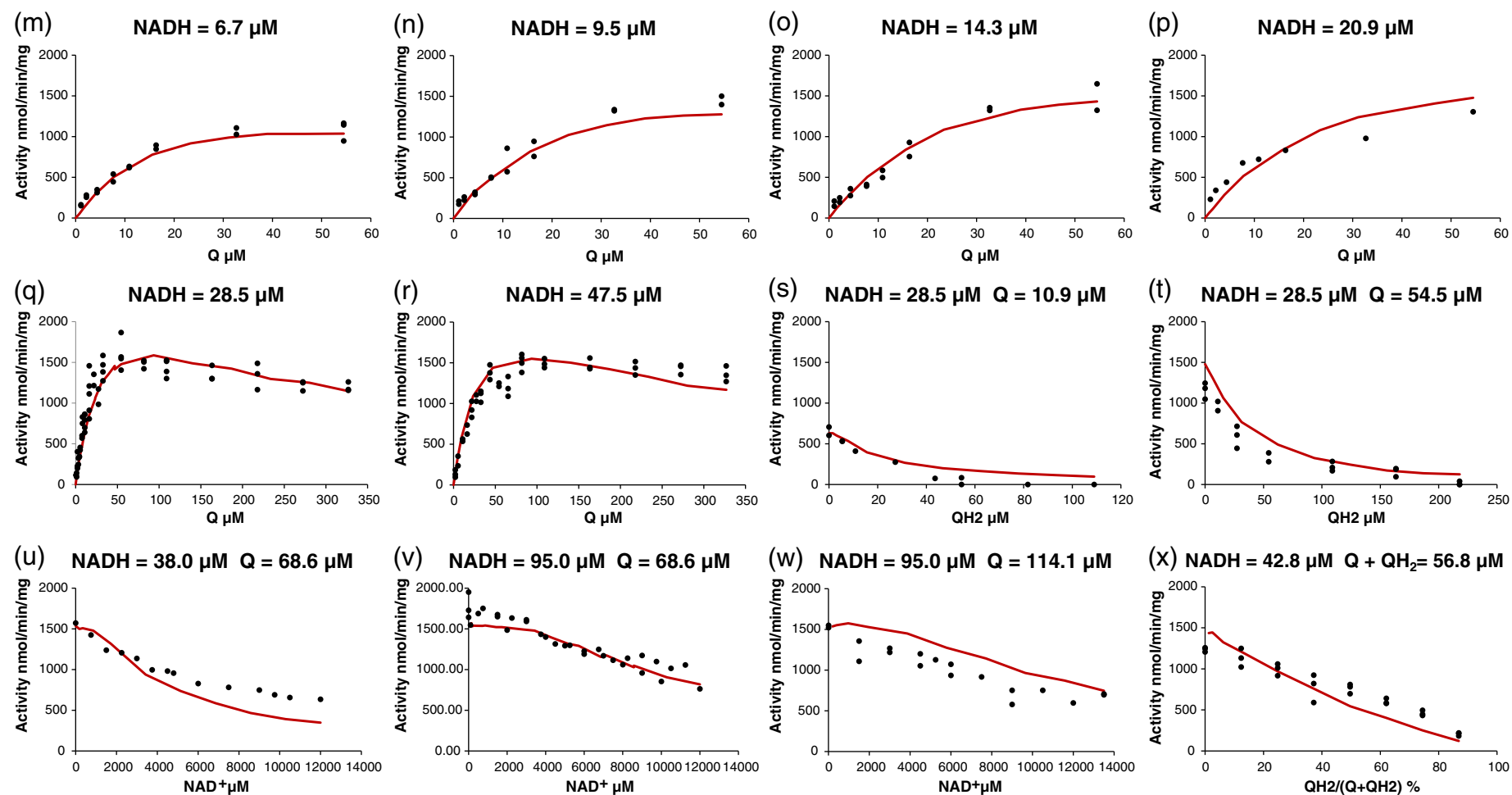


Fig. 5 (continued).

is also clear on the simulated curve according to our simple model described in [29] which validates the pertinence of the simulation curves (Fig. 3C). Moreover, the plateau is more pronounced at low concentration of Q. Such a plateau is not in accordance with a classical ordered or ping-pong mechanism. This behavior is due to the fact that the electron uptake by Q is a limiting step in regards to the reaction $\text{NADH} \Rightarrow \text{NAD}$ which gives the electrons to the FeS centers. It is experimentally well known that the reduction of ferricyanide by NADH involving only flavin is at least an order of magnitude higher than the maximal rate of CoQ reduction [40]. Thus, at high NADH concentrations, the rate of the global reaction cannot increase with NADH concentration and the FeS centers become more and more reduced (Fig. 6). When the same points/simulations are plotted as a function of Q a plateau is also observed but only at low concentrations of NADH (Figs. 2 and 3B and D) because it is only in this condition that NADH electron transfer to the chain of FeS centers is limiting. With low NADH and saturating Q, the FeS centers are more oxidized as shown in Fig. 6. When the NADH concentration is increased, the plateau disappears (Fig. 3B and D) because the reaction $\text{Q} \Rightarrow \text{QH}_2$ becomes limiting and the points and curves in Fig. 2(n), (o) and (p) are quasi identical at high Q concentrations. Thus the observation of a plateau principally as a function of NADH concentration is a strong indication that the two redox reactions at both ends of the pathway ($\text{NADH} \Rightarrow \text{NAD}$ and $\text{Q} \Rightarrow \text{QH}_2$) are independent and that the electron transfer to Q molecules i.e. the reaction $\text{Q} \Rightarrow \text{QH}_2$ is most of the time the limiting process, with a buffering accumulation of electron in between the redox reactions. This correlates with the rate constants of the electron transfer from NADH to FMN and vice versa which are in the order of 10^{11} while the electron transfers from the N2 center to Q or SQ is of the order of 10^5 .

However, we do not completely reject the ping-pong mechanism. It should be stressed that our first analysis of kinetic data concluded to a ping-pong mechanism. However, the reciprocal plots gave non linear plots in most cases, well evidenced even at low concentration, with scattered data Fig. 3. This deviation from a ping-pong mechanism might be explained by experimental uncertainties, the extent of subtracted rotenone inhibition (around 5%) or the presence of the membrane potential. However, this behavior was retrieved in the simulation process which does not take into account the proton motive force. This is a good indication that the observation of a plateau was neither an artifact nor due to the membrane potential, but an actual characteristic of complex I mechanism, which can be easily explained by the electron buffering due to the FeS clusters chain. As a matter of fact, a ping-pong mechanism is a good approximation of the experimental results and probably holds at low NADH concentration and moderate Q concentrations (between 30 and 70 μM) for which the rates of NADH oxidation and ubiquinone reduction are of the same order of magnitude, preventing electron accumulation in the

intermediate redox centers (see Fig. 6). Lower Q concentration however, leads to electron accumulation which blocks NADH oxidation, giving the plateau profile. It should be noticed that, in the direct plot, the deviation from the hyperbolic behavior is weak which renders the ping-pong mechanism usable when modeling complex I in a global model of respiratory chain (paper in preparation). As mentioned in the Materials, methods and models section, we also used a hyperbolic fitting of the individual curves (with an inhibitory term when necessary) to help us in the fitting procedure. Further information can be derived from these fittings. The secondary plots of the $K_{m(\text{app})}$ and $V_{\text{max}(\text{app})}$ of each hyperbole fitting Fig. 5 (and Fig. 2) curves, plotted as function of both substrates are also hyperbolic functions that have similar $Q_{0.5}$ and $\text{NADH}_{0.5}$: $Q_{0.5} = 28$ and $26 \mu\text{M}$; $\text{NADH}_{0.5} = 7.7$ and 5.1 for $K_{m(\text{app})}$ and $V_{\text{max}(\text{app})}$ curves respectively. This result is in accordance with the equation of a ping-pong mechanism and is an additional reason to consider this mechanism as a good approximation of the experimental results. Furthermore, the asymptotic values of these secondary plots give another way to estimate $Q_{0.5}$ and $\text{NADH}_{0.5}$ ($31 \mu\text{M}$ and $10.1 \mu\text{M}$, respectively) which are in agreement with the previous values. They also permit to estimate the maximum (V_{max}) of the $V_{\text{max}(\text{app})}$: 2726 and 2653 nmol/min/mg deduced from the $V_{\text{max}(\text{app})}$ curves as a function of Q and NADH respectively. The fact that the V_{max} value is not reached experimentally illustrates once more the deviation from a ping-pong mechanism due to the existence of a plateau.

4.4. Inhibitions by the products

An inhibition by the products (NAD^+ and QH_2) is evidenced with half-inhibition around $2000 \mu\text{M}$ and $25 \mu\text{M}$ respectively which are slightly higher than the K_d determined in Table 3 ($682 \mu\text{M}$ and $8.5 \mu\text{M}$). The binding/release rate constants in Table 4 appear appropriate to fit these inhibitions curves (Fig. 5(s) and (t) (QH_2) and (u) to (w) (NAD^+)). It should be stressed here that a strong inhibition by decylubiquinol was already reported as impeding complex I assay [9].

4.5. Inhibitions by quinone

In addition, experimental kinetics at high concentration of the substrate ubiquinone, evidenced an inhibition which was previously reported [6,10]. In its simple form described in [29] our model did not predict any inhibition by this substrate. It is thus necessary to consider another (inhibitory) ubiquinone binding site. A second ubiquinone site was already proposed in complex I for kinetic reasons [10], in a model of inversed Q cycle (discussed in Sherwood and Hirst [41]) or in EPR spectroscopy observation [11]. However, the recent structures of complex I [13–15] do not afford any evidence for

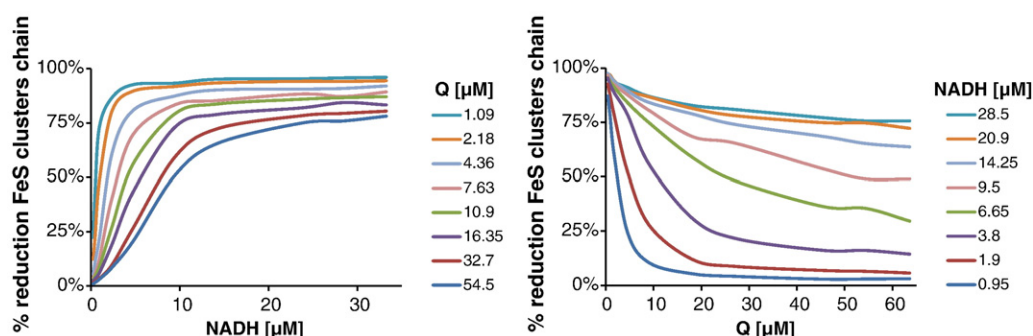


Fig. 6. Simulations of the percentage of reduction of the FeS clusters at different substrates concentrations. Model 1 was used to calculate these values. Model 2 gives very similar results.

the existence of a second Q site and at this stage, the existence of a second Q site on complex I is unlikely.

The recent complex I structures [14,15] nevertheless brought new features which have to be taken into account in the kinetic modeling of complex I: first a broad ubiquinone binding pocket which accommodates also all complex I inhibitors and second a possible quinone exchange path leading from cluster N2, the supposed site of ubiquinone reduction, to the N terminal beta sheet of the 49 kDa subunit (in the membrane). The case of *Yarrowia lipolytica* is worth to be mentioned with the quinone pocket being far away from the membrane. This led to the hypothesis of a hydrophobic ramp which could link the ubiquinone pocket to the membrane [15]. In the case of *T. thermophilus* [14,27] the distance between the membrane and the quinone site around N2 is much shorter, but nevertheless appreciable (25 Å) and also necessitates some movement of the quinone out of the membrane. Although there is no evidence of a precise second ubiquinone site, there is the possibility of ubiquinone taking several positions either in the large reaction pocket or on the way leading from the membrane to the reaction site and thus impeding or hindering the access of the ubiquinone substrate by ubiquinol or the release of the ubiquinol product by ubiquinone.

We modeled this diffusion of quinone towards or from the reaction site by a second quinone site, site B, situated on an obligatory path of the ubiquinone towards the reaction site (site A close to N2) or of the ubiquinol from the reaction site towards its release. The model is represented on Fig. 4. In this model the “ideal” path is represented by the central line for which a ubiquinone molecule first binds to site B then pass to site A, where it is transformed in ubiquinol and released after its back passage through site B. However, competitions occur at site B and can hinder the access of Q or the release of QH₂, thus leading to inhibition of the reaction. We would like to stress once more that the incorporation in the model of a quinone inhibitory site is not the proof of the existence of a second site for quinone but is a way to model the congestion of a quinone path or of a large pocket around the reaction site.

As shown in Fig. 5 it is possible to find a set of parameters (Tables 4 and 5) which reasonably fit our kinetic data performed in several conditions especially at high concentrations of substrates and products. Let us emphasize once more that the parameters of our model are not all adjustable. All the internal rate constants are imposed by the structure and the redox properties of the redox centers.

The K_d for NADH and NAD is similar to those of the previous model with only one Q site: 24 μM as compared to 9.8 μM for NADH and 643 μM as compared to 682 μM for NAD.

In a recent paper, Wikström and Hummer [5] gave a mechanism very similar to the one we propose here with two quinone/quinol sites QA and QB (see Fig. 2 of their paper), except that, in their model, the quinone/quinol molecules do not move from one site to the other but the electrons are exchanged between the quinone/quinol bound to both sites. From a modeling point of view, this is not very different. It must be noticed however that their model will not easily explain the inhibition of complex I by high concentration of quinone.

4.6. Thermodynamical considerations and transconformations

It has been proposed that the apparent complex kinetic mechanism could be the result of transconformational mechanisms and certainly change(s) of conformation are necessary to couple the oxido-reduction reaction to the transfer of protons. For instance, Chen et al. report in [42] that a model taking into account a transconformation linked to the proton extrusion lead to “a hybrid ping-pong rapid-equilibrium random bi-bi mechanism, consolidating the characteristics from previously reported kinetic mechanisms and data”. A pseudo-reversible active/inactive transition was reported for a long time ago by Vinogradov [28]. The slow transitions $k_{A \rightarrow B}$ and $k_{B \rightarrow A}$ between sites A and B (Table 5) which appear in the fitting of our

experimental data could well be the indication of transconformational changes. Traces of the energy of transconformation may also be found in a thermodynamical analysis of our binding results particularly at N2 site. At this site we can write: $\Delta G(N2 \rightarrow Q_f) = EbQ + \Delta G(N2 \rightarrow Q_b) - EbQH_2$, where Q_f and Q_b means Q free and Q bound respectively, EbQ and $EbQH_2$ are the binding energy of Q and QH₂ which are calculated (in V) from the K_d in Tables 3 and 4 (last column). The calculation of $\Delta G(N2 \rightarrow Q_f)$ from the midpoint potential, taking $EmN2 = -100$ mV and $EmQH_2/Q(\text{free}) = +45$ mV (Table 1) gives $\Delta G(N2 \rightarrow Q_f) = -290$ mV (for 2 electrons). We calculated $\Delta G(N2 \rightarrow Q_b) = -80$ mV from Table 1 ($Em Q/QH_2 \text{ bound} = -60$ mV). EbQ (-272 mV) and $EbQH_2$ (-312 mV) will never fill in the difference ($EbQ + \Delta G(N2 \rightarrow Q_b) - EbQH_2 = -40$ mV with these values). This is a strong indication that something else happens. There is a large consensus in the literature that admits that the energy of vectorial protons transduction has to be gained from some point of the oxido-reduction reactions through a conformational change of the protein. The only place to gain sufficient energy to accomplish this process is the transition $2 N2_{red} + Q \rightarrow 2 N2_{ox} + QH_2$ [5,11,43]. This could explain the discrepancy between the two calculations, because the transconformation energy as well as protons binding are not taken into account in the balance equation above. The same discrepancy exists in NADH binding, but to a lesser extent, and may be explained by the release and binding of protons. It reinforces the idea [5,11,43] that the energy necessary for protons extrusion is gained in the reduction of ubiquinone.

4.7. Interest of a stochastic model

In conclusion, we would like to emphasize the interest of using a stochastic model. First it allows us to easily express the structural and physical properties of the redox centers inside a single, isolated complex I (more generally inside any single respiratory complexes). The kinetic properties are the expression of the structure of complex I with its physical (redox) characteristics. Because it is based on the probabilities of reactions it mimics rather closely what occurs *in vivo* to the electrons inside a single complex I. The history of each electron can be followed during a simulation. Second, it is easy to accumulate simulations and to take their mean to simulate what experimentally occurs in a spectrophotometer cuvette as we did here. It offers a simple way to a bottom up approach of enzymatic kinetic.

A detailed scheme of how electron tunnel along the chain of seven FeS clusters from FMN to N2 was recently published by Hayashi and Stuchebrukhov [44]. They evidenced different tunneling pathways, the role of “some key residues for electron transfer and demonstrated the essential role played by internal water at the interface between protein subunits as a mediator of electron transfer”. Our approach does not go into these details. It takes the empirical equations of Moser and Dutton expressing the rate constants (and thus the probabilities) of electron transition between neighbor redox centers as a function of the distance between the redox centers and their midpoint potentials which represents a kind of average of all electron paths described in [45]. It should be stressed that in our model we do not consider the electrostatic interaction between neighbor redox centers, which could affect the electron transfer according to simple coulombic laws. The group of Stuchebrukhov, on theoretical bases, determined the intrinsic redox potential of FeS centers of complex I from experimental titration curves, taking into account the coulombic interactions between the centers [46] and evidenced a rollercoaster profile. They also gave the extent of the coulombic interactions for different values of the relative permittivity $\epsilon_r = 5$ or 20 [47]. At this stage of our model, we have not taken into account these effects which could influence the kinetics and the apparent mechanism derived from the simulation. Using our present model we can calculate the percentage of reduced centers at a given steady state (Fig. 6) for variable NADH or Q concentrations with fixed concentrations of the other substrate. It is clear from these curves that

this percentage is variable according to the substrates concentrations. Thus the coulombic effect will depend on substrates concentrations. A more detailed model is underway in our lab to account for these changes in midpoint potentials of the FeS centers according to their variable charged environment. It will not fundamentally change the conclusions of our paper because the k_{et} is more sensitive to the distances (D) than to the ΔG^0 . Furthermore; in all cases, the numerous redox centers in between the two half of the global reaction necessarily introduce a buffer effect separating both half of the redox reaction whatever the influences of the electrostatic interactions are.

We are now waiting for new structural reports that would be invaluable to incorporate the coupling of redox reactions to the extrusion of protons in our model. In its present state we think that our model, emphasizing the structural and functional separation of the two half of the global reaction, gives a realistic representation of the kinetic behavior of NADH–Ubiquinone oxidoreductase, in the absence of a proton gradient.

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